A Two-Stage Approach to Quantifying NBA Scoring Talent

Alex Susi
Georgia Institute of Technology

Introduction

The NBA is often referred to as a "make-or-miss" league. However, not all makes nor misses are created equal. A contested layup, an open midrange pull-up, and a wide open catch-and-shoot 3-pointer all have different expected conversion rates.

Shot Type	Defense	Dribbles	N	FG%
Rim	Tight	2	4,770	65.6
Mid-range	Open	1	2,407	45.7
3-point	Wide Open	0	35,293	38.3

Table 1: Summary table by shot type, defender distance, and dribbles.

Typical metrics such as FG%, eFG%, and TS% conflate **shot quality** with **shooting skill**.

We introduce **Expected Points Above Average (EPAA)**, a framework that:

1. Estimates *shot quality* using contextual features.
2. Estimates *shooting talent* via Bayesian hierarchical modeling.

By controlling for shot quality, closest defender, and team defense, **EPAA identifies the most proficient scorers with full posterior uncertainty**, allowing front offices and fans to more accurately assess scoring abilities.

This work uses public NBA tracking data from over 200,000 field goal attempts during the 2024–25 regular season, accessed via the `hoopR` package.

Methodology

Stage 1: Shot Quality

"How hard was this shot for an average player guarded by an average defender?"

- Expected Field Goal Percentage (xFG) is estimated using three XGBoost classifiers (one per shot type: rim, mid-range, 3-point), trained on play-by-play features.
- Player and defender identities are intentionally excluded in this stage.

Stage 2: Shooting Talent

"How good is each player at converting shots after controlling for shot quality?"

- Modeled using a Bayesian hierarchical framework in Stan with random effects for shooters, individual defenders, and defensive teams.
- Partial pooling shrinks lower-volume players toward the league mean while preserving separation for high-volume stars, producing skill parameters with uncertainty quantification rather than simple point estimates.

Code

GitHub repo: `github.com/alex-susi/exp-fg-pct`

Pipeline: R + Stan (`cmdstanr`)

Artifacts: Code, models, graphs, and output CSVs.

Data: NBA Stats API via `hoopR`

Contact: `alex.susi2298@gmail.com`

Stage 1: Shot Quality

Training the model across the 2024-25 season yields an xFG for every shot attempt.

The SHAP plots highlight each feature's importance to xFG. A positive SHAP value corresponds to an increase in xFG. Feature importance varies across shot types.

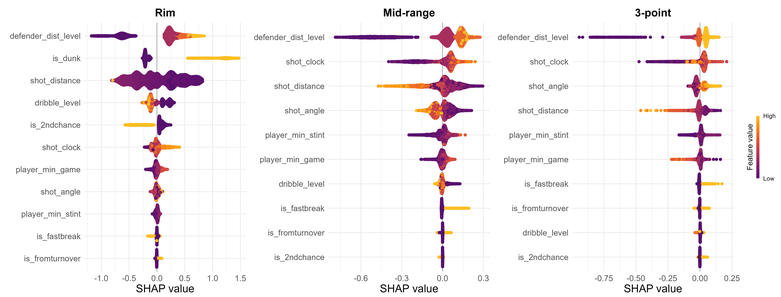

Figure 1: **SHAP Feature Importance by Shot Type**. For rim attempts, defender distance and dunks explain most of the variability. Defender distance, shot clock, shot distance, and shot angle rank near the top for mid-range and 3-pointers.

Outputs from the Shot Quality model are transformed on the logit scale and used as a baseline in the Shooting Talent model.

Stage 2: Shooting Talent

For each shot attempt $n = 1, \ldots, N$, let:

$$y_n \in \{0, 1\} \quad \text{shot outcome (make/miss)}$$
$$t_n \in \{1, 2, 3\} \quad \text{shot type (rim, mid-range, three-point)}$$

Likelihood and Linear Predictor

$$y_n \sim \text{Bernoulli}\left(\text{logit}^{-1}(\eta_n)\right)$$
$$\eta_n = \underbrace{\alpha_{t_n} + \beta_{t_n} \cdot \text{xfg_logit}_n}_{\text{Shot Quality}} + \underbrace{a^{\text{p}}_{j[n],\,t_n}}_{\text{shooter talent}} + \underbrace{a^{\text{f}}_{d[n],\,t_n}}_{\text{individual defender}} + \underbrace{a^{\text{m}}_{k[n],\,t_n}}_{\text{team defense}}$$

Random effects jointly model shooter skill, individual defender ability, and team defensive scheme — three sources typically confounded in shooting metrics.

Random Effects

$$a^{\text{p}}_{j,t} = \mu^{\text{vol}}_t \cdot x^{\text{vol}}_{j,t} + \sigma^{\text{p}}_t \cdot z^{\text{p}}_{j,t}, \quad z^{\text{p}}_{j,t} \sim \mathcal{N}(0,\ 1) \text{ for all players } j = 1, \ldots, J$$
$$a^{\text{f}}_{d,t} = \sigma^{\text{f}}_t \cdot z^{\text{f}}_{d,t}, \quad z^{\text{f}}_{d,t} \sim \mathcal{N}(0,\ 1) \text{ for all defenders } d = 1, \ldots, D$$
$$a^{\text{m}}_{k,t} = \sigma^{\text{m}}_t \cdot z^{\text{m}}_{k,t}, \quad z^{\text{m}}_{k,t} \sim \mathcal{N}(0,\ 1) \text{ for all teams } k = 1, \ldots, K$$

where $x^{\text{vol}}_{j,t}$ is standardized $\log(1 + \text{FGA per 36 min})$ for player j in shot type t.

Priors

$$\alpha_t \sim \mathcal{N}(0,\ 0.25^2) \quad \beta_t \sim \mathcal{N}^{+}(1,\ 0.25^2)$$
$$\mu^{\text{vol}}_t \sim \mathcal{N}(0,\ 0.3^2)$$
$$\sigma^{\text{p}}_t \sim \mathcal{N}^{+}\left(0,\ s^{\text{p}}_t\right), \quad s^{\text{p}} = (0.30,\ 0.30,\ 0.25)$$
$$\sigma^{\text{f}}_t \sim \mathcal{N}^{+}\left(0,\ s^{\text{f}}_t\right), \quad s^{\text{f}} = (0.20,\ 0.20,\ 0.20)$$
$$\sigma^{\text{m}}_t \sim \mathcal{N}^{+}\left(0,\ s^{\text{m}}_t\right), \quad s^{\text{m}} = (0.12,\ 0.10,\ 0.06)$$

Results

Player Leaderboards

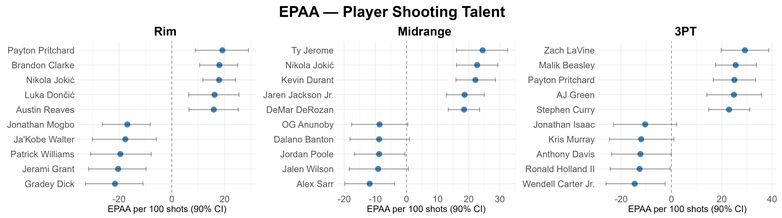

Figure 2: **Top 5 and Bottom 5 Players by EPAA per 100 shots**

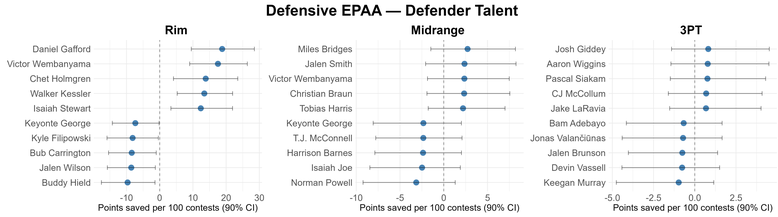

Figure 3: **Top 5 and Bottom 5 Defenders by Defensive EPAA per 100 shots contested**

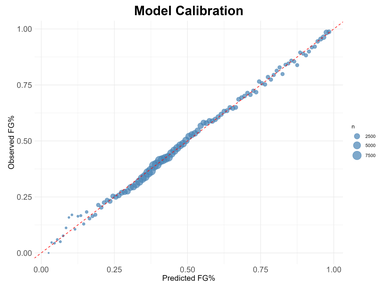

Figure 4: **Model Calibration**. Tight adherence to the 45-degree line confirms model calibration.

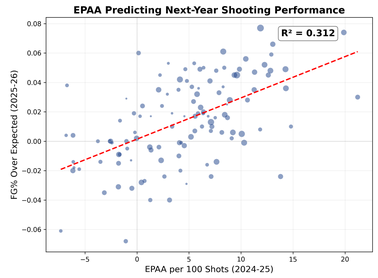

Figure 5: **Year-over-Year Predictive Power** of EPAA for FG% over expected.

Key Findings

- EPAA rewards efficiency while penalizing inflated stats from easier shots.
- Defender rankings reveal individual rim protection skill has tighter credible intervals than mid-range and 3-point defense (possibly more luck-driven).
- Credible intervals for all defensive team parameters crossed 0.

Conclusion

EPAA demonstrates that a two-stage framework can **separate shot difficulty from shooting skill** with principled uncertainty quantification. The $R^2 = 0.31$ year-over-year correlation shows EPAA captures stable shooting talent signal.

Limitations & Future Direction

- The shot quality models are limited by the public NBA API. Detailed player tracking data could provide richer features and more accurate xFG estimations.
- Future work will train models over multiple seasons of data.
- Incorporating foul-drawing value would better capture a player's scoring impact.